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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
December 2025

BT2103-1 – MOLECULAR BIOLOGY AND GENETIC ENGINEERING
(For BTY)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. The transforming principle identified by Avery, MacLeod, and McCarty was destroyed by:
A) Proteases
B) RNases
C) DNases
D) Lipases
2. In the Hershey-Chase experiment, ³⁵S labeled phage showed radioactivity only in the supernatant after infection. This indicated that:
A) Proteins entered the bacterial cell
B) DNA entered the bacterial cell
C) Both proteins and DNA entered the bacterial cell
D) RNA was the genetic material
3. Euchromatin regions are expected to be enriched in:
A) a) Satellite DNA repeats
B) Actively transcribed genes
C) Inactive centromeric sequences
D) Heterochromatic proteins
4. In human karyotyping, banding patterns obtained by Giemsa staining help in:
A) Detecting repetitive DNA sequences
B) Locating ribosomal RNA genes
C) Identifying individual chromosomes and structural abnormalities
D) Measuring telomere length
5. Retroviruses use reverse transcriptase to:
A) Produce double-stranded RNA from RNA
B) Repair damaged RNA molecules
C) Insert histones during replication
D) Produce complementary DNA (cDNA) from RNA
6. The Okazaki fragments are produced because:
A) DNA polymerase can synthesize in both directions
B) DNA polymerase can synthesize only in the 5'→3' direction
C) Helicase pauses on the lagging strand
D) Replication starts from multiple points simultaneously
7. Human chromosomes are classified based on:
A) Length and gene content
B) Centromere position
C) Telomere structure
D) Nucleosome organization
8. Euchromatin is generally
A) Loosely packed and transcriptionally active
B) Highly condensed and transcriptionally inactive
C) Found only in prokaryotes
D) Made only of RNA



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9. The core bacterial RNA polymerase consists of:
 - A) $\alpha 2\beta\beta'\omega$
 - B) $\alpha\beta\gamma\delta$
 - C) $\alpha 2\beta\delta\sigma$
 - D) $\alpha\beta\beta'\sigma$
10. The role of the sigma (σ) factor in prokaryotic transcription is:
 - A) Catalysis of phosphodiester bond formation
 - B) Termination of transcription
 - C) Recognition of promoter sequences
 - D) Proofreading errors
11. In prokaryotes, transcription termination that requires rho protein is called:
 - A) Intrinsic termination
 - B) Rho-independent termination
 - C) Factor-independent termination
 - D) Rho-dependent termination
12. The 5' cap added to eukaryotic mRNA consists of:
 - A) Methylated guanosine triphosphate
 - B) Poly-A tail
 - C) Inverted uracil base
 - D) tRNA molecule
13. RNA splicing in eukaryotes is carried out by:
 - A) Exonucleases
 - B) Spliceosome complex
 - C) RNA polymerase II
 - D) tRNA synthetase
14. Which codon does not code for an amino acid and functions as a stop codon?
 - A) AUG
 - B) UGG
 - C) GUG
 - D) UAA
15. Restriction enzymes were first discovered in:
 - A) Bacteria
 - B) Fungi
 - C) Viruses
 - D) Plants
16. Type II restriction enzymes are most widely used in recombinant DNA technology because:
 - A) They cut at random sites
 - B) They modify DNA
 - C) They degrade RNA
 - D) They cut DNA at specific recognition sequences
17. A selectable marker gene in a cloning vector usually confers:
 - A) Resistance to antibiotics
 - B) Ability to degrade toxins
 - C) Fluorescent properties
 - D) Increased plasmid copy number
18. Calcium chloride treatment in prokaryotic transformation works by:
 - A) Strengthening DNA strands
 - B) Neutralizing negative charges on DNA and cell membranes
 - C) Breaking phosphodiester bonds
 - D) Digesting cell walls
19. Liposome-mediated transfer is effective because:
 - A) Liposomes protect DNA and fuse with cell membranes
 - B) Liposomes degrade cellular RNA
 - C) Liposomes produce DNA ligase
 - D) Liposomes block restriction enzymes
20. The microprojectile or biolistic method delivers DNA by:
 - A) Chemical uptake of plasmids
 - B) Bombardment of cells with DNA-coated particles
 - C) Liposome-mediated uptake
 - D) Use of bacteriophage vectors

PART - B: DESCRIPTIVE ANSWER QUESTIONS

	Unit – I	Marks	BT*	CO*	PO*
1. a)	Explain the Central Dogma of molecular biology. How is this concept modified by the discovery of reverse transcription?	8	L*1	1	1,2
b)	Compare the structure and functions of DNA polymerases in prokaryotes and eukaryotes.	8	L2	2	1,2
2. a)	Describe Avery, MacLeod, and McCarty's experiment. How did it help to establish DNA as the genetic material?	8	L1	1	1,2

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|-------|---|---|----|---|-----|
| b) | Discuss the different types of DNA damages. Add a note on its consequences. | 8 | L2 | 2 | 1,2 |
| 3. a) | Discuss the experimental proof for semiconservative mode of DNA replication. | 8 | L1 | 1 | 1,2 |
| b) | With a neat labelled diagram, discuss in detail the components in DNA replication fork. | 8 | L1 | 2 | 1,2 |

Unit – II

- | | | | | | |
|-------|--|---|----|---|------|
| 4. a) | Write an essay on the properties of the genetic code. Why is it said to be universal, degenerate, and non-overlapping? | 8 | L1 | 4 | 2,6 |
| b) | Discuss transcriptional control of gene expression in eukaryotes, focusing on promoters, enhancers, and silencers. | 8 | L2 | 3 | 2,12 |
| 5. a) | Describe the process and significance of post-translational modifications in proteins. | 8 | L1 | 3 | 2,12 |
| b) | Compare regulation of gene expression in prokaryotes versus eukaryotes. | 8 | L2 | 4 | 2,6 |
| 6. a) | Describe the steps of protein synthesis in prokaryotes: initiation, elongation, and termination. | 8 | L1 | 3 | 2,6 |
| b) | Explain the lac operon as a model for inducible gene regulation in prokaryotes. | 8 | L1 | 4 | 2,12 |

Unit – III

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|-------|--|---|----|---|------|
| 7. a) | Give a detailed account on DNA-modifying enzymes (ligases, polymerases, phosphatases, kinases). Add a note on their role in molecular cloning. | 8 | L1 | 5 | 2,12 |
| b) | Describe plasmids as vectors in recombinant DNA technology. Add a note on their essential features. | 8 | L1 | 5 | 2,12 |
| 8. a) | Explain the biology of the Ti plasmid of <i>Agrobacterium tumefaciens</i> and its role in plant genetic engineering. | 8 | L1 | 5 | 2,12 |
| b) | Give a detailed account on physical modes of gene transfer. Add a note on the respective advantages and disadvantages. | 8 | L1 | 5 | 2,12 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
